



LumenCore

Evidence engineering for AI, data, software, and technical decisions

Mission fit

Independent review, acquisition evidence, data provenance, model and workflow evaluation, and controlled prototype-to-pilot decisions.

Best-fit role

A specialized evidence-engineering workstream under a qualified prime or a bounded review sprint using public or buyer-authorized data.

Core Capabilities

- Source hashing and schema-aware ingestion
- Frozen evaluation plans and baseline-versus-candidate replay
- Machine-readable claim and evidence states
- Append-only artifact and decision receipts
- Negative, neutral, inconclusive, and adverse-result retention
- Human approval gates for promotion, rerun, hold, or rejection
- Reviewer-facing reproducibility and conformance packages

Why This Is Useful to a Government Reviewer

Review stage	Control delivered
Before evaluation	Lock the source, owner, baseline, metric, threshold, holdout, failure rule, and decision authority.
During execution	Record code, configuration, environment, inputs, outputs, exceptions, deviations, and hashes.
After execution	Separate measured evidence from assumptions; preserve adverse and inconclusive results; retain reviewer decisions.
At promotion	Require an authorized human decision to promote, rerun, hold, reject, or seek independent review.

Bounded Engagement Model

Evidence-readiness sprint: agree on one decision, one authorized data bundle, one incumbent baseline, one candidate workflow, frozen metrics, named failure conditions, and a reviewer-owned acceptance state. Deliver the reproducible evidence bundle and limitations without production connectivity.

Deliverables and Reviewer Answers

A concise package designed to answer what was tested, against what, under which authority, with which failures retained, and what the evidence does and does not authorize.

Representative Deliverables

- Source and authority registry
- Reproducible execution bundle
- Evidence-to-claim traceability matrix
- Locked evaluation and acceptance plan
- Deviation and failure log
- Reviewer decision packet with explicit limitations

Evidence-to-Decision Workflow

Step	Gate	Reviewer-visible record
1	Register	Sources, ownership, authority, access, freshness, and identity
2	Freeze	Baseline, candidate, metric, threshold, holdout, exclusions, and failure rules
3	Execute	Exact software, configuration, environment, inputs, outputs, and deviations
4	Review	Results, uncertainty, negative controls, adverse cases, and unresolved gaps
5	Authorize	Human promote, rerun, hold, reject, or independent-review decision

Procurement and Teaming Posture

Prime boundary: LumenCore does not represent current federal past performance, a production ATO, FedRAMP authorization, CMMC certification, a facility clearance, cleared personnel, agency endorsement, or field validation.

Teaming boundary: Security, domain licenses, controlled-data access, key-person qualifications, staffing, pricing, representations, and submission authority remain with the qualified prime and Government.

Relevant routing categories: custom software and data services, computer systems design, engineering and R&D; support, technical evaluation, and acquisition evidence. Confirm the controlling NAICS, PSC, set-aside, and entity record for each notice.

Fast Reviewer Questions

What is proven now? Present now: reusable source, replay, receipt, conformance, and human-gate software patterns. Snapshot 2026-07-29: 126 direct comparisons (120 exploratory; 6 confirmatory), 0 internal promotion passes, and 0 global Holm positives.	What is not proven? Agency deployment, operational suitability, field performance, certification, clearance, endorsement, or realized savings.
What can start safely? A bounded public or buyer-authorized data review with a frozen plan and no production control. Named protocol snapshot 2026-08-04: 0 eligible future observations; waiting for new source rows.	Who decides? The authorized Government reviewer, buyer, or qualified prime. LumenCore does not automate promotion authority.

Contact and Action Boundary

Public repository: <https://github.com/robertashworth1986-debug/lumen-core-public>

Public domain: <https://lumen-core.ai> (verify availability at action time; use the repository as the stable code entry point).

Identifiers: use current reviewed company and SAM records; private identifiers are omitted.

External use: requires a current notice check, source-bound claim and duplicate review, and authorized human approval.